Practice Sheet 1

Part A

1. Perform each of the indicated operations:

$i)\ (i-2)\{2(1+i)-3(i-1)\}$ $ii)\ \frac{(2+i)(3-2i)(1-i)}{(1-i)^2}$ $iii)\ (2i-1)^2\left(\frac{4}{1-i}+\frac{2-i}{1+i}\right)$

$iv)\ 3\left(\frac{1+i}{1-i}\right)^2-2\left(\frac{1-i}{1+i}\right)^3$ $v)\ \frac{3i^{10}-i^{19}}{2i-1}$ $vi)\ \frac{i^4-i^9+i^{16}}{2-i^5+i^{10}-i^{15}}$

2. Show that, $(5+3i)+\{(-1+2i)+(7-5i)\}=\{(5+3i)+(-1+2i)\}+(7-5i)$ illustrates the associative law of addition.

3. If $z_1=1-i,\quad z_2=-2+4i,\quad z_3=\sqrt{3}-2i,$ evaluate each of the following:

(i) $|2z_2-3z_1|^2$ (ii) $\left|\frac{z_1+z_2+1}{z_1-z_2+i}\right|$ (iii) $\overline{(z_2+z_3)(z_1-z_3)}$ (iv) $\mathrm{Re}\{2z_1^3+3z_2^3-5z_3^2\}$

(v) $\mathrm{Im}\left\{\frac{z_1z_2}{z_3}\right\}$ (vi) $z_1^2+2z_1-3$ (vii) $\left|z_1\overline{z_2}+z_2\overline{z_1}\right|$ (viii) $\frac{1}{2}\left(\frac{z_3}{\overline{z_3}}+\frac{\overline{z_3}}{z_3}\right)$

4. Express each of the following complex number in polar form and show them graphically.

(i) $2+2\sqrt{3}\,i$ (ii) $2\sqrt{2}+2\sqrt{2}\,i$ (iii) $-2\sqrt{3}-2i$ (iv) $-1+\sqrt{3}\,i$

5. Prove that: (i) $\overline{z_1+z_2}=\overline{z_1}+\overline{z_2}$ (ii) $|z_1z_2|=|z_1||z_2|$ (iii) $|z_1\pm z_2|\le|z_1|+|z_2|$

(iv) $|z_1\pm z_2|\ge|z_1|-|z_2|$.

6. State and prove De Moivre's Theorem.
7. Evaluate each of the following by De Moivre's Theorem:

$i)\ \frac{(8\ cis\ 40°)^3}{(2\ cis\ 60°)^4}$ $ii)\ \frac{\left(3e^{i\frac{\pi}{6}}\right)\left(2e^{-i5\frac{\pi}{4}}\right)\left(6e^{5i\frac{\pi}{3}}\right)}{\left(4e^{2i\frac{\pi}{3}}\right)^2}$ $iii)\ (5\ cis\ 20°)(3\ cis\ 40°)$ $iv)\ (2\ cis\ 50°)^6$

8. Find all the roots of the following equations.

$i)\ z^3=(-1+i)$ $ii)\ z^5=(-4+4i)$ $iii)\ z^4=-16i$ $iv)\ z^6=64i$ $v)\ z^4+z^2+1=0$

Part B

1. Perform the indicated operations analytically and graphically.

   $i)\ (2 + 3i) + (4 - 5i)$ $\qquad$ $ii)\ (7 + i) - (4 - 2i)$

2. Describe geometrically the set of points z satisfying the following conditions:

   $i)\ Re(z) > 1$ $\qquad$ $ii)\ |2z + 3| > 4$
   $iii)\ 1 < |z - 2i| < 2$ $\qquad$ $iv)\ |z + 1 - i| \leq |z - 1 + i|$
   $v)\ Re(z) \geq 0$ $\qquad$ $vi)\ Im(z) \geq 0$
   $vii)\ |z - 4| \geq |z|$ $\qquad$ $viii)\ |z - 2| \leq |z + 2|$
   $ix)\ Re\left(\frac{1}{z}\right) \leq 1/2$ $\qquad$ $x)\ \frac{\pi}{2} < \arg arg\ z < \frac{3\pi}{2},\ |z| > 2$

3. Using the properties of conjugate and modulus, show that:
   i. $\overline{\bar{z} + 3i} = z - 3i$
   ii. $\overline{iz} = -i\bar{z}$
   iii. $\left|(2\bar{z} + 5)\left(\sqrt{2} - i\right)\right| = \sqrt{3}\,|2z + 5|$
   iv. $|2z + 3\bar{z}| \leq 4|Re(z)| + |z|$

4. Find the modulus and argument of the following complex numbers:

   $i)\ \frac{2-i}{2+i}$ $\qquad$ $ii)\ \frac{\sqrt{3}+i}{\sqrt{3}-i}$

   $iii)\ \left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^2$

5. Prove that $|z - i| = |z + i|$ represents a straight line.
6. Prove that $|z + 2i| + |z - 2i| = 6$ represents an ellipse.
7. Find an equation of a circle center at $(2, 3)$ with radius 3.
8. Sketch the region in xy-plane represented by the following set of points:

   $i)\ Re(\bar{z} - 1) = 2$ $\qquad$ $ii)\ Im\left(z^2\right) = 4$
   $iii)\ \left|\frac{2z-3}{2z+3}\right| = 1$ $\qquad$ $iv)\ Re(z) + Im(z) = 0$